task2

August 7, 2026

[73]:
```
import pandas as pd

data = pd.read_csv("Social_Network_Ads.csv")
data = data.drop("User ID", axis=1)
print(data.head())
print(data.shape)
data.dropna(inplace=True)
data.shape
```

```
   Gender  Age  EstimatedSalary  Purchased
0    Male   19            19000          0
1    Male   35            20000          0
2  Female   26            43000          0
3  Female   27            57000          0
4    Male   19            76000          0
(400, 4)
```

[73]:
```
(400, 4)
```

[74]:
```
features = ["Gender", "Age", "EstimatedSalary"]
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(data[features],␣
 ↪data["Purchased"], test_size=0.2)
```

[75]:
```
X_train_encoded = pd.get_dummies(X_train, columns=["Gender"], drop_first=True)
X_test_encoded = pd.get_dummies(X_test, columns=["Gender"], drop_first=True)

from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()

X_train_encoded[["Age", "EstimatedSalary"]] = scaler.
 ↪fit_transform(X_train_encoded[["Age", "EstimatedSalary"]])
X_test_encoded[["Age", "EstimatedSalary"]] = scaler.
 ↪transform(X_test_encoded[["Age", "EstimatedSalary"]])

print(X_train_encoded.head())
```

```
          Age  EstimatedSalary  Gender_Male
90  -1.536536         0.313077         True
39  -1.057864        -1.139296        False
83  -0.291990         0.516409         True
86  -1.345067        -0.442157         True
204  1.909900         0.894026        False
```

[76]:
```
from sklearn.naive_bayes import GaussianNB

model = GaussianNB()
model.fit(X_train_encoded, y_train)
```

[76]:
```
GaussianNB()
```

[77]:
```
y_pred = model.predict(X_test_encoded)
```

[83]:
```
from sklearn.metrics import accuracy_score, classification_report
print(y_pred)
print(y_test.to_numpy())
print("Accuracy: ", accuracy_score(y_test, y_pred))
print("Classification: ", classification_report(y_test, y_pred))
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 1 0 0 1 1 0 1 0 0 1 1 0 0 0 0 1 0 1 0
 0 0 0 0 1 1 0 0 0 1 1 0 0 0 0 0 1 0 1 0 1 0 0 0 1 0 0 1 0 1 0 0 1 0 0 0 0
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 0 0 0 0 1 1 0 0 0 0 1 0 0 1 0 0 1 0 1 0 1 0 0 0 0 0 0 0 0 1 1 1 0 1 1 0 0 0
 0 0 0 1 0 0]
Accuracy:  0.8125
Classification:                precision    recall  f1-score   support

           0       0.82      0.92      0.87        53
           1       0.80      0.59      0.68        27

    accuracy                           0.81        80
   macro avg       0.81      0.76      0.77        80
weighted avg       0.81      0.81      0.80        80
```